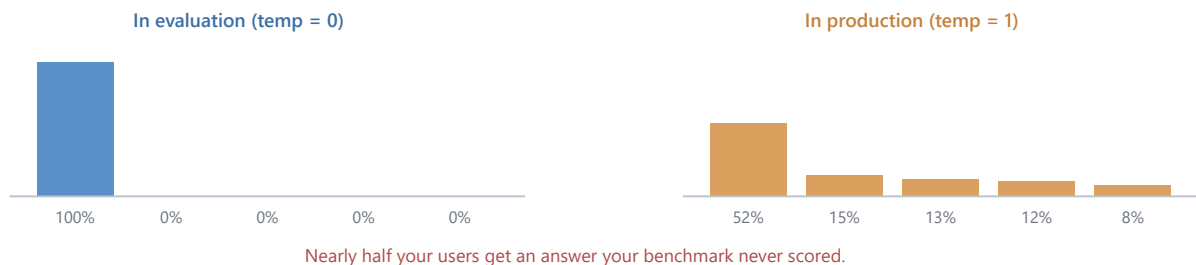


1. A mean hides a distribution

Evaluated at temperature 0, a model gives the same answer every time — 100% consistency. In production at temperature 1, the same prompt might give the top answer 52% of the time and four other answers the rest. Same mean score, wildly different user experience.

And on current frontier models you can't even buy the determinism. The classic version of this problem is a discipline failure: someone evaluated at temperature 0 and shipped at temperature 1. But as we saw in Part 1, several frontier models no longer accept a temperature at all — passing one is an error. You cannot pin evaluation to a deterministic point even if you want to, because there is no longer a knob that produces one. The single number your eval reports is a *sample* from a distribution, in the lab exactly as in production. So "a mean hides a distribution" stops being a mistake you can avoid by being careful, and becomes the permanent condition of measuring these systems: run each eval prompt several times and look at the spread, because the spread is the product.



Nearly half your users get an answer your benchmark never scored.

The benchmark measured one point. Production samples the whole distribution.

2. Real prompts are nothing like benchmark prompts

Benchmark prompt	Real user prompt
"Write a function in Python that returns the factorial of a number."	"ok python factorial function pls for homework, it needs to handle negative numbers too i think?? idk the hw doesnt really say but my ta mentioned it"

3. Long conversations change behaviour

Benchmarks are usually single-turn. Real conversations run for hours. Past some length, models drift — tone changes, instructions from the system prompt fade against thousands of tokens of chat, and behaviour gets strange. This is not hypothetical; it's the well-documented failure mode behind several very public chatbot incidents.

4. Users ignore intended use

Whatever you designed the system for, someone will use it for something else — prompt injection, roleplay, therapy, homework, jailbreaks. Your evaluation set contains none of it.